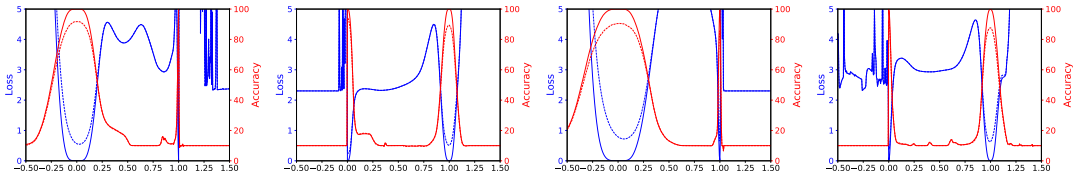
A.3 Small-Batch vs Large-Batch for ResNet-56

Similar to the observations made in Section 5, the "sharp vs flat dilemma" also applies to ResNet-56 as shown in Figure 14. The generalization error for each solution is shown in Table 1. The 1D and 2D visualizations with filter normalized directions are shown in Figure 15.

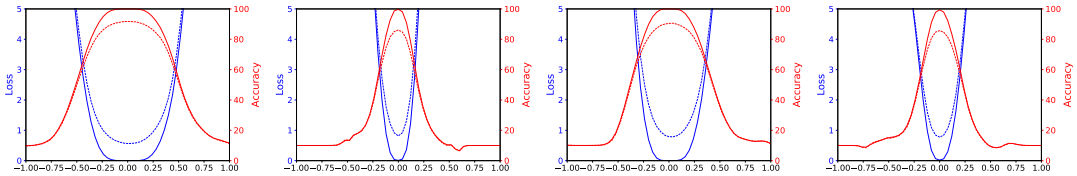
Table 1: Test errors for ResNet-56 with different optimizer, batch-size and weight-decay.

	SGD		Adam	
	bs=128	bs=4096	bs=128	bs=4096
WD = 0	8.26	13.93	9.55	14.30
WD = 5e-4	**5.89**	**10.59**	**7.67**	**12.36**

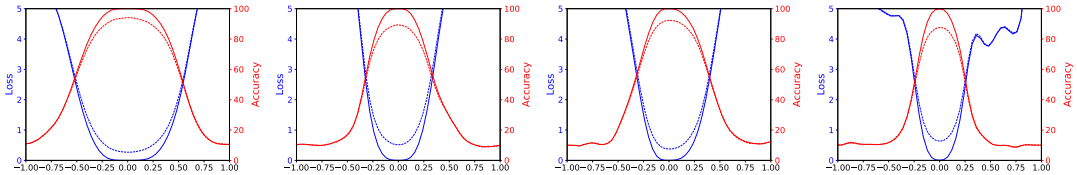


(a) SGD, WD=0 (b) SGD, WD=5e-4 (c) Adam, WD=0 (d) Adam, WD=5e-4

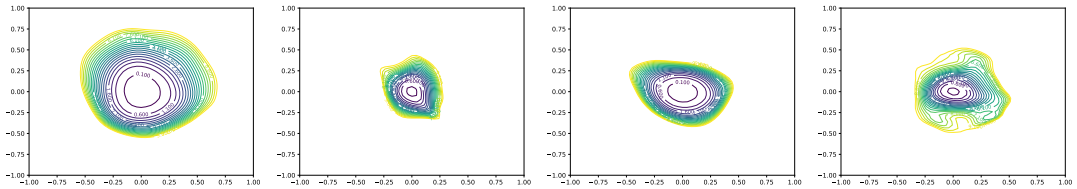
Figure 14: 1D linear interpolation of solutions obtained by small-batch and large-batch methods for ResNet-56. The blue lines are loss values and the red lines are error.



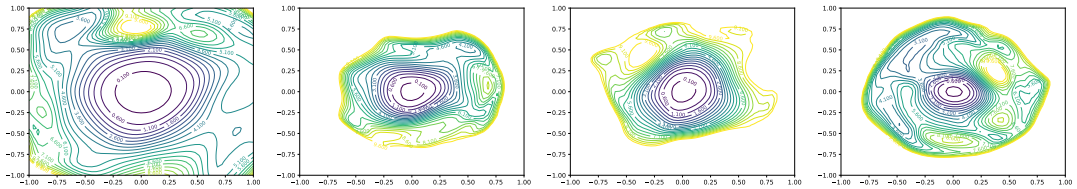
(a) SGD, 128, 8.26% (b) SGD, 4096, 13.93% (c) Adam, 128, 9.55% (d) Adam, 4096, 14.30%



(e) SGD, 128, 5.89% (f) SGD, 4096, 10.59% (g) Adam, 128, 7.67% (h) Adam, 4096, 12.36%



(i) SGD, 128, 8.26% (j) SGD, 4096, 13.93% (k) Adam, 128, 9.55% (l) Adam, 4096, 14.30%

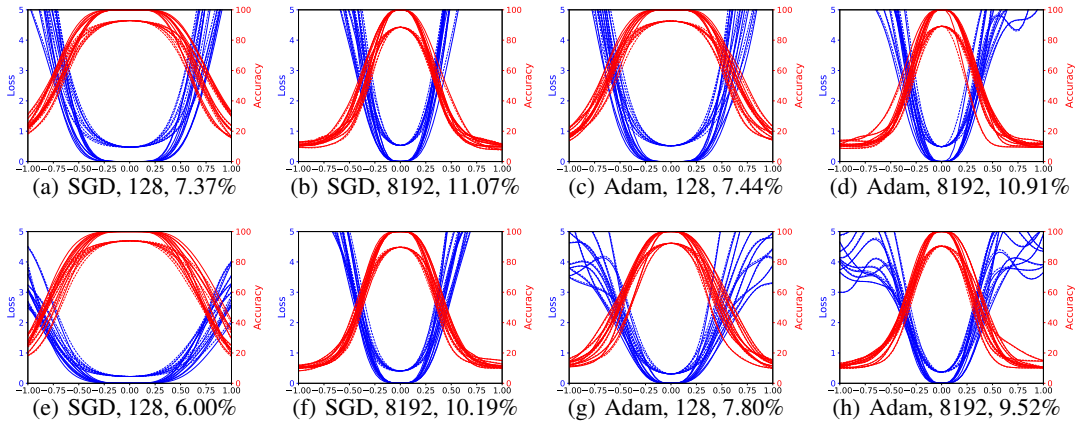


(m) SGD, 128, 5.89% (n) SGD, 4096, 10.59% (o) Adam, 128, 7.67% (p) Adam, 4096, 12.36%

Figure 15: 1D and 2D visualization of ResNet-56 trained with different optimizer, batch size, and weight decay. The first and third row uses zero weight decay and the second and fourth row uses 5e-4 weight decay.

A.4 Repeatability of the Loss Surface Visualization

Do different random directions produce dramatically different plots? We plot the 1D loss surface of VGG-9 with 10 random filter-normalized directions. As shown in Figure 16, the plots are very close in shape. We also repeat the 2D loss surface plots multiple times for ResNet-56-noshort, which has worse generalization error. As shown in Figure 17, there are apparent changes in the loss surface for different plots, however, the qualitative choatic behaviour is quite consistent across plots.



(a) SGD, 128, 7.37% (b) SGD, 8192, 11.07% (c) Adam, 128, 7.44% (d) Adam, 8192, 10.91%

(e) SGD, 128, 6.00% (f) SGD, 8192, 10.19% (g) Adam, 128, 7.80% (h) Adam, 8192, 9.52%

Figure 16: Repeatability of the surface plots for VGG-9 with filter normalization. The shape of minima obtained using 10 different random filter-normalized directions.

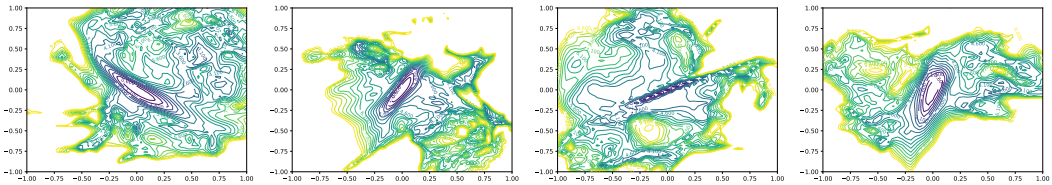


Figure 17: Repeatability of the 2D surface plots for ResNet-56-noshort. The model is trained with batch size 128, initial learning rate 0.1 and weight decay 5e-4. The final training loss is 0.192, the training error is 6.49 and the test error is 13.31.

A.5 Implementation Details

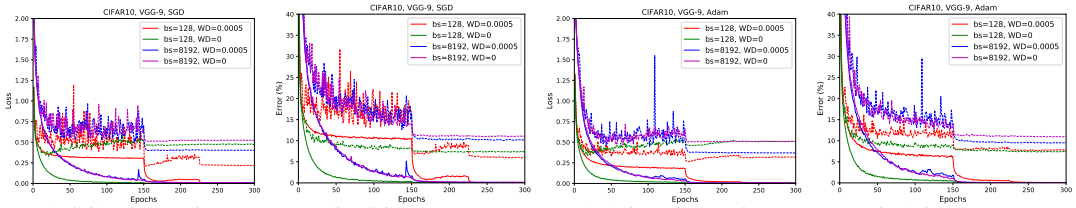
Computing resources for generating the figures Our PyTorch code can be executed in a multiple GPU workstation as well as an HPC with hundreds of GPUs using `mpi4py`. The computation time depends on the model's inference speed on the training set, the resolution of the plots, and the number of GPUs. The resolution for the 1D plots in Figure 3 is 401×401. The default resolutions used for the 2D contours in Figure 3 and Figure 5 is 51×51. We use higher resolutions (251×251) for the ResNet-56-noshort used in Figure 1 to show more details. For reference, a 2D contour plot of ResNet-56 with a (relatively low) resolution of 51×51 will take about 1 hour on a workstation with 4 GPUs (Titan X Pascal or 1080 Ti).

Batch Normalization parameters In the 1D linear interpolation methods, the Batch Normalization (BN) parameters including the "running mean" and "running variance" need to be considered as part of θ. If these parameters are not considered, then it is not possible to reproduce the exact loss values for both minimizers. In the filter-normalized visualization, the random direction perturbs all weights except batch norm parameters. Note that the filter normalization process removes the effect of weight scaling, and so the batch normalization can be ignored.

The VGG-9 architecture and parameters for Adam VGG-9 is a cropped version of VGG-16, which keeps the first 7 Conv layers in VGG-16 with 2 FC layers. A BN layer is added after each conv layer and the first FC layer. We find VGG-9 is an efficient network with better performance comparing to VGG-16 on CIFAR-10. We use the default values for β_1, β_2 and ϵ in Adam with the same learning rate schedule as used in SGD.

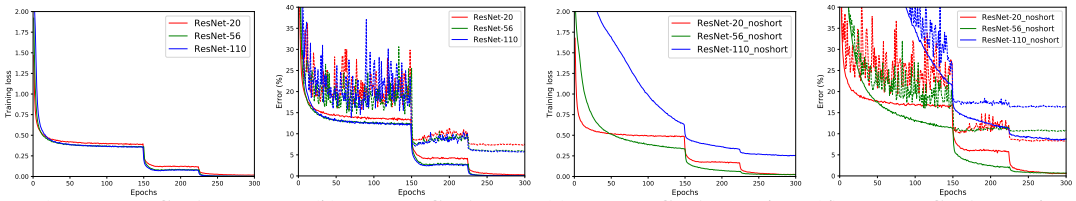
A.6 Training Curves for VGG-9 and ResNets

The loss curves for training VGG-9 used in Section 5 are shown in Figure 18. Figure 19 shows the loss curves and error curves of architectures used in Section 6 and Table 2 shows the final error and loss values. The default setting for training is using SGD with Nesterov momentum, batch-size 128, and 0.0005 weight decay for 300 epochs. The default learning rate was initialized at 0.1, and decreased by a factor of 10 at epochs 150, 225 and 275.



(a) SGD, loss values (b) SGD, errors (c) Adam, loss values (d) Adam, errors

Figure 18: Training loss/error curves for VGG-9 with different optimization methods. Dashed lines are for testing, solid for training.



(a) ResNet-CIFAR (b) ResNet-CIFAR (c) ResNet-CIFAR-noshort (d) ResNet-CIFAR-noshort

Figure 19: Convergence curves for different architectures.

Table 2: Loss values and errors for different architectures trained on CIFAR-10.

	init LR	Training Loss	Training Error	Test Error
ResNet-20	0.1	0.017	0.286	7.37
ResNet-20-noshort	0.1	0.025	0.560	8.18
ResNet-56	0.1	0.004	0.052	5.89
ResNet-56-noshort	0.1	0.192	6.494	13.31
ResNet-56-noshort	0.01	0.024	0.704	10.83
ResNet-110	0.1	0.002	0.042	5.79
ResNet-110-noshort	0.01	0.258	8.732	16.44